

Jacob Kavanal

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Education

University of California - San Diego *BS, Data Science*

Sept 2024 - March 2027 — GPA: 3.81

San Jose State University *BS, Computer Science*

Aug 2022 - May 2024 — GPA: 3.89

Experience

Data Engineering Intern *ORiON*

Danville, CA — Sept 2025 - Present

- Designed and implemented **data processing pipelines** that handle **~25k listings/day** using **Python** and **Playwright**, preparing data for downstream ML applications.
- Deployed scalable solutions onto **AWS Lambda** (Cloud Computing) with **API Gateway**, automating data capture and persisting results in a **MongoDB** database.
- Tested **React (Next.js 14)** components for an AI job platform using **Jest**, ensuring software reliability for integrated systems.

Market Manager & Software Instructor *CodeAdvantage*

San Diego, CA — Dec 2023 - Present

- Rapidly promoted (Instructor → Tech Manager → Market Manager) to direct technical operations, client relations, and curriculum delivery for the San Diego market.
- Standardized coding curricula for **Python and JavaScript**, breaking down complex CS fundamentals (Object-Oriented Programming, control flow, data structures) into scalable training modules.
- Managed software licensing and hardware infrastructure for **30+ partner schools**, establishing technical SOPs that ensured 100% uptime for daily coding classes.
- Led the recruitment and technical training of new instructors, ensuring consistent teaching quality and adherence to engineering best practices across the region.

Web Design Intern *San Jose Works*

San Jose, CA — Nov 2021 - Dec 2022

- Led team of three to develop website for the San Jose Works Department, ensuring strict adherence to design specifications.
- Wrote **custom scripts** in **JS** to automate Excel processing of **1,200** applications, reducing analysis time by **75%**.
- Met and presented technical work to representatives of **Mayor Sam Liccardo**.

Projects

Mixify [🔗](#)

January 2026

- Developed a **feature engineering pipeline** to analyze **real-time audio signals** (tempo, key, loudness) via **Spotify Web API** for a recommendation system.
- Built a custom algorithm for **signal compatibility** using BPM (± 5 range) and harmonics to automate track selection.
- Implemented **OAuth 2.0** with **PKCE flow** for secure authentication, ensuring robust integration between external APIs and the React frontend.
- Deployed to **Vercel**, utilizing environment variables for secure configuration management in a cloud-native environment.

Recipe Data Analysis

June 2025

- Performed **Exploratory Data Analysis (EDA)** and **Statistical Modeling** using **NumPy** and **pandas** to uncover patterns in seasonal nutrition data.
- Visualized complex dataset trends using **Seaborn** and **matplotlib**, generating insights for further model development.
- Trained and evaluated models with **5-fold CV**, achieving **$R^2=0.78$** and **MAE=42.6 kcal** with inference time **12 ms/sample**.
- Assessed model fairness with custom hypothesis testing workflows implemented in Python using **SciPy** and **scikit-learn**.

Cult Classic Calculator

April 2025

- Collaboratively developed a full-stack **Machine Learning application** to predict movie success probabilities based on user-defined attributes.
- Designed a **data processing pipeline** using **FastAPI** and **Pydantic** to feed features into a trained decision tree model.
- Utilized **scikit-learn** for training and evaluation, achieving **AUC=0.81** and **Acc=76%** through weighted factor analysis.

Tweet Sentiment Analysis

December 2023

- Built a **Multinomial Naive Bayes Model** using **scikit-learn**, achieving **96.5% accuracy** on a dataset of **2,000 tweets**.
- Implemented an advanced **NLP preprocessing pipeline** (tokenization, negation handling, TF-IDF vectorization) to clean unstructured text data.
- Containerized the application for deployment, demonstrating familiarity with software packaging and scalable architecture.
- Developed a real-time web interface using **React + Node.js** to visualize sentiment classification results.

Organizations: SJSU Software Engineering Society (AI/ML Development Team), Data Science Students Society (Consulting)

Technologies: Python, Pandas, NumPy, **PyTorch**, **Scikit-learn**, **AWS (Cloud)**, **Git**, Signal Processing, SQL, React, Node.js, MongoDB, Linux/Unix, Jest

Coursework: **Linear Algebra**, **Probability & Statistics**, Principles of Data Science, Intro to **Machine Learning**, Data Structures and Algorithms, Computer Architecture, Theoretical Foundations of Data Science